

Specialized Pathology Question

How can bone scintigraphy differentiate prosthesis loosening from infection in artificial joint replacement?



Why relevance ranking fails for evidence retrieval

Relevance $\neq$ Supportiveness

Aboutness-oriented retrieval

Supportiveness-aware reranking



Dense retrieval



*Lexicon-guided
sparse retrieval*

Weaknesses:

*X Optimizes topical similarity,
not evidence*

*X Terminology variation hurts
matching*

*X Non-supportive passages
occupy top ranks*

Strengths:

✓ High recall / reachability

✓ Efficient candidate generation

✓ Robust to some paraphrases



*Supportive Evidence
Verifier (SEV)*

Weaknesses:

*X Limited by first-stage recall
(needs reachable evidence)*

*X Requires supervision / verifier
training data*

*X Threshold/score calibration
needed*

Strengths:

*✓ Targets answer-justifying
evidence*

*✓ Improves top-ranked
supportiveness*

✓ Works within fixed top-k1 pool

Reranking

Downstream QA

Answer generation